AI-Powered Parameter Optimization for Graphene Metamaterial THz Devices

FYP Progress Report - Phase 1: Data Analysis

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Date: March 2026

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1. Project Objectives

This project aims to develop a Machine Learning tool that predicts and optimises the performance of graphene-based terahertz (THz) metamaterial modulators. These devices are simulated in COMSOL Multiphysics, where each simulation takes approximately 24 hours. The goal is to replace this brute-force approach with an ML model that can predict device performance instantly.

Primary Objectives

- **Objective 1:** Maximise S12/S21 dip depth:

The S12 parameter measures signal transmission through the device. A deeper dip (more negative dB value) indicates stronger resonance. Current best is -13.41 dB; the target is -20 dB or deeper.

- **Objective 2:** Maximise frequency shift between ON and OFF states:

The device operates in two states controlled by graphene conductivity (sigma=0.3 mS for ON, sigma=1.2 mS for OFF). The frequency at which the S12 dip occurs shifts between these states. A larger shift means better tunability. Current best is 60 GHz.

The 7 Tuneable Parameters

Parameter	Description	Current Range	Unit
g_w	Capacitor gap width	3 - 5	um
c_w	Capacitor width	12 - 28	um
dx	Unit cell size	23 - 35	um
w_au	Gold line width	4 (fixed)	um
w_au_top	Top gate bar thickness	1.5 (fixed)	um
w_graph	Graphene extra width	1 (fixed)	um
h_graph	Graphene extra height	2 (fixed)	um

2. Methodology - What Was Done in This Phase

In this initial phase, the focus was on building the data processing pipeline and performing exploratory data analysis on the available COMSOL simulation results.

Steps Completed

- **Data Collection:** Received 13 unique COMSOL simulation data files from supervisor, each containing S12 and S22 parameter curves across a frequency range of 300-800 GHz.
- **Data Cleaning:** Identified and removed 12 duplicate files (Windows download copies). Cleaned dataset contains 13 unique simulations.
- **Pipeline Development:** Built a Python data processing pipeline (data_loader.py) that automatically parses COMSOL export files, extracts simulation parameters from filenames, identifies S12 dips, and pairs ON/OFF states.
- **Visualisation:** Created 5 analysis plots (visualize.py) examining parameter effects on S12 dip depth and frequency shift.
- **Analysis:** Generated summary statistics and parameter sensitivity analysis (analyze.py) to understand which parameters have the strongest influence on device performance.

3. Dataset Overview

The dataset consists of 13 unique COMSOL simulations. Each file contains S-parameter data (S12 transmission and S22 reflection) across 26 frequency points from 300 GHz to 800 GHz. The simulations vary three geometric parameters (dx, g_w, c_w) and the graphene conductivity (sigma).

Complete Simulation Table

#	dx (um)	g_w (um)	c_w (um)	sigma (mS)	S12 Dip (dB)	Dip Freq (GHz)
1	23	3	28	0.3	-11.08	540
2	25	3	28	0.3	-11.35	500
3	27	3	28	0.3	-11.65	480
4	31	3	28	0.3	-12.08	440
5	33	3	28	0.3	-12.31	420
6	35	3	28	0.3	-12.48	400
7	35	3	28	1.2	-13.41	360
8	35	5	12	0.3	-12.00	600
9	35	5	12	1.2	-9.61	540
10	35	5	20	0.3	-12.17	520
11	35	5	20	1.2	-11.20	460
12	35	5	28	0.3	-12.64	460
13	35	5	28	1.2	-12.48	400

ON/OFF Pair Summary

Out of 13 simulations, 4 complete ON/OFF pairs were identified (matching geometric parameters with both sigma=0.3 and sigma=1.2).

dx	g_w	c_w	ON Dip (dB)	ON Freq (GHz)	OFF Dip (dB)	OFF Freq (GHz)	Freq Shift (GHz)
35	3	28	-12.48	400	-13.41	360	40
35	5	12	-12.00	600	-9.61	540	60
35	5	20	-12.17	520	-11.20	460	60
35	5	28	-12.64	460	-12.48	400	60

4. Graph Analysis and Findings

4.1 All S12 Transmission Curves

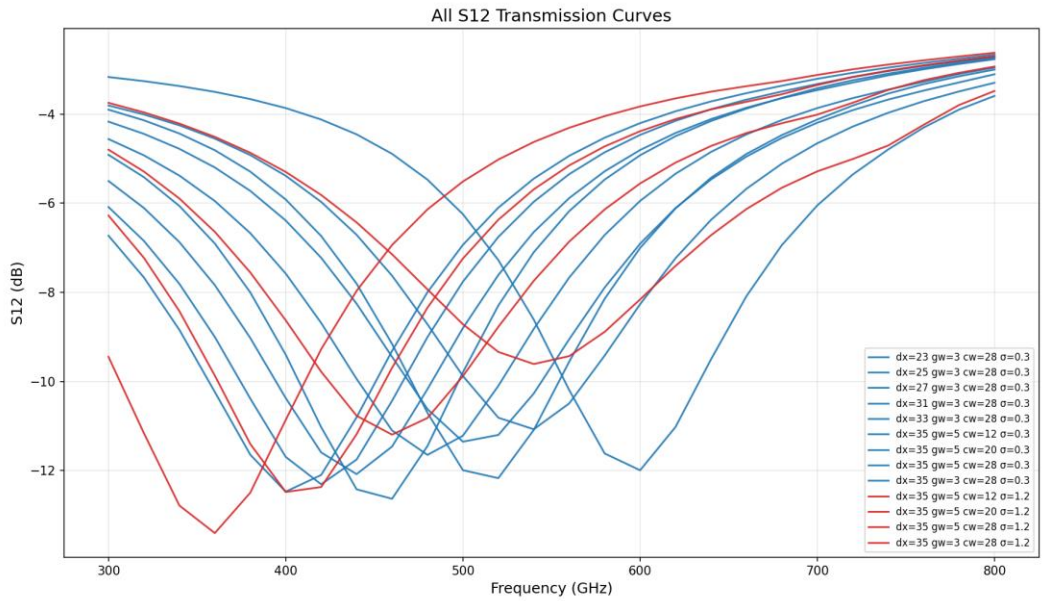

Figure 1: All S12 transmission curves. Blue = ON state (sigma=0.3), Red = OFF state (sigma=1.2)

Figure 1 shows all 13 S12 transmission curves overlaid. The blue curves represent the ON state (graphene conductivity sigma=0.3 mS) and the red curves represent the OFF state (sigma=1.2 mS). Key observations:

- Each curve shows a clear resonance dip where transmission is minimised, indicating strong electromagnetic absorption by the metamaterial structure.
- The dip frequency shifts depending on the geometric parameters - larger capacitor width (c_w) and larger unit cell size (dx) push the resonance to lower frequencies.
- The ON and OFF states show visibly different dip positions, confirming that the graphene conductivity effectively tunes the resonance frequency.
- All dips fall in the range of -9.6 to -13.4 dB, still significantly above the -20 dB target.

4.2 Effect of Capacitor Width (c_w) on S12 Dip Depth

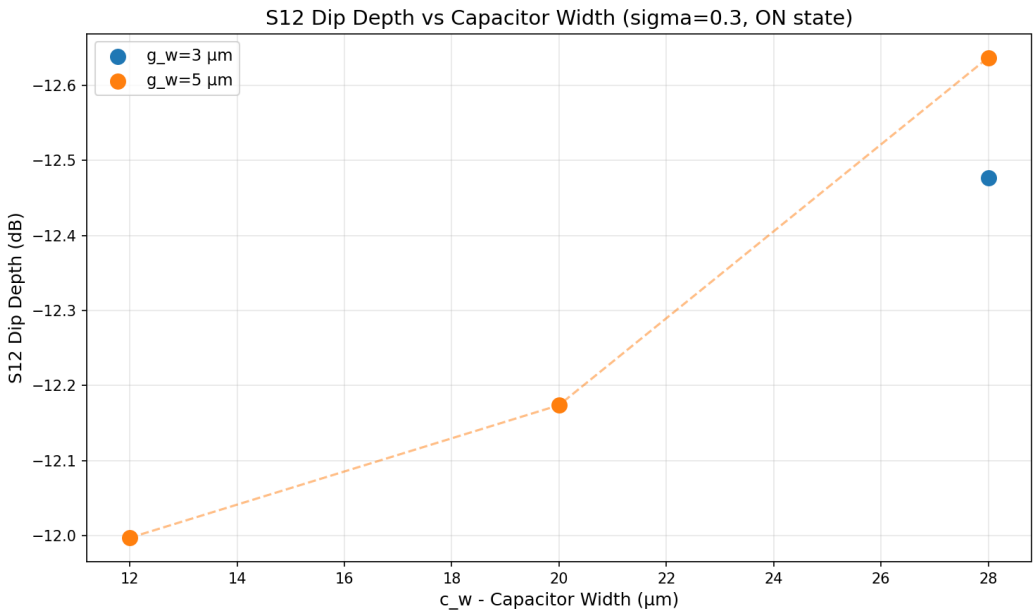

Figure 2: S12 dip depth vs capacitor width (c_w) for g_w=3 and g_w=5 at dx=35

Figure 2 isolates the effect of capacitor width (c_w) on the S12 dip depth, comparing two gap width values (g_w=3 and g_w=5). Analysis:

- Increasing c_w from 12 to 28 um deepens the S12 dip for g_w=5 (from -12.0 to -12.6 dB), showing a clear positive correlation between capacitor width and resonance strength.
- The g_w=5 configuration (orange) consistently shows deeper dips than g_w=3 (blue) at c_w=28, suggesting that a wider capacitor gap improves resonance coupling.
- The sensitivity is approximately 0.040 dB per um of c_w increase - a moderate effect.
- Extrapolating this trend, even c_w=50 um would only reach approximately -13.5 dB, suggesting that c_w alone cannot achieve the -20 dB target. Other parameters must be explored.

4.3 Effect of Unit Cell Size (dx) on S12 Dip Depth

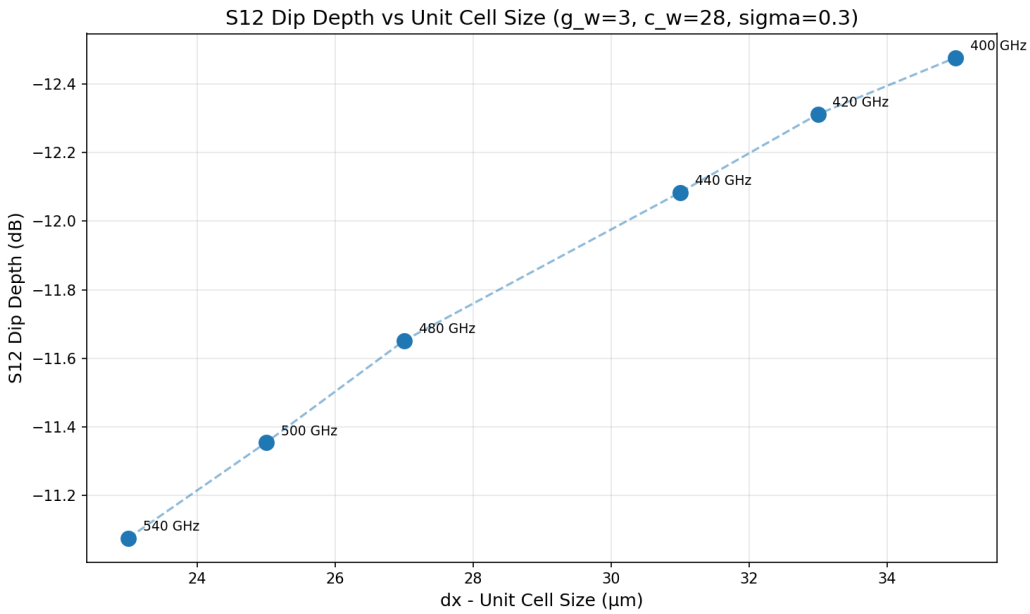

Figure 3: S12 dip depth vs unit cell size (dx) for g_w=3, c_w=28

Figure 3 shows the dx sweep data - six simulations where only the unit cell size varies from 23 to 35 um. This reveals a strong linear relationship:

- Increasing dx from 23 to 35 um deepens the S12 dip from -11.1 to -12.5 dB, a change of 1.4 dB over 12 um.
- The sensitivity is 0.117 dB per um - approximately 3x stronger than the c_w effect. This makes dx the most influential parameter identified so far.
- The resonance frequency also shifts significantly: from 540 GHz (dx=23) down to 400 GHz (dx=35), a shift of 140 GHz across the sweep range (11.7 GHz per um).
- The near-linear trend suggests that further increasing dx beyond 35 um could yield even deeper dips, though physical constraints may limit this.

4.4 ON vs OFF State Comparison

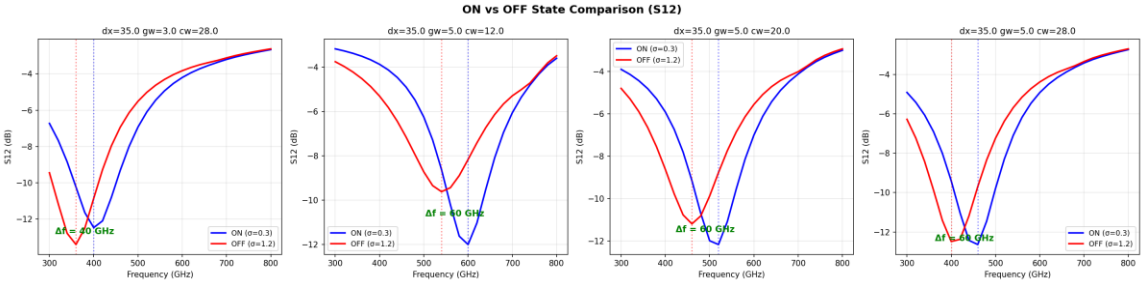

Figure 4: ON (blue) vs OFF (red) state S12 curves for all 4 complete pairs

Figure 4 shows side-by-side comparisons of the 4 complete ON/OFF pairs. The green annotation shows the frequency shift between states:

- Three configurations with g_w=5 show a consistent 60 GHz frequency shift between ON and OFF states, regardless of c_w value (12, 20, or 28 um). This suggests that the ON/OFF shift is primarily controlled by the graphene properties, not the capacitor geometry.
- The g_w=3, c_w=28 configuration shows a smaller shift of only 40 GHz, indicating that smaller gap width reduces the graphene's ability to tune the resonance.
- In the OFF state (sigma=1.2), the dips tend to be shallower (less negative) except for the g_w=3, c_w=28 case where the OFF dip (-13.4 dB) is actually deeper than the ON dip (-12.5 dB).
- The g_w=5, c_w=28 configuration offers the best trade-off: 60 GHz shift with the deepest average dip (-12.6 dB).

4.5 Frequency Shift and Dip Depth Summary

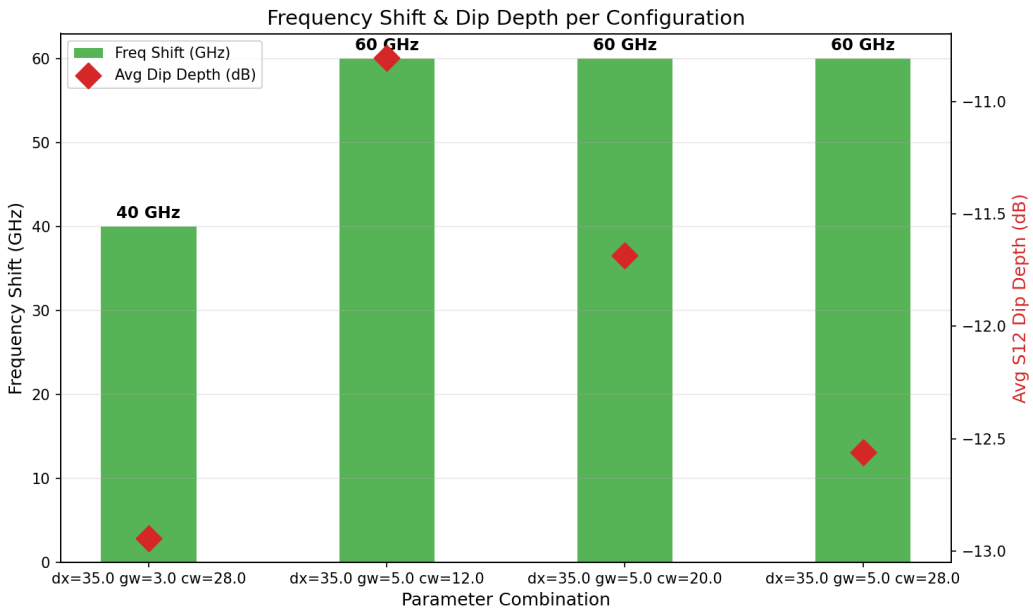

Figure 5: Frequency shift (bars) and average dip depth (diamonds) per configuration

Figure 5 combines both optimisation objectives in a single chart. The green bars show the frequency shift, while the red diamonds show the average dip depth:

- The three g_w=5 configurations all achieve 60 GHz frequency shift (the maximum observed), while the g_w=3 configuration only achieves 40 GHz.
- Among the 60 GHz shift configurations, g_w=5, c_w=28 has the deepest average dip (-12.6 dB), making it the best overall configuration found so far.
- There is a clear trade-off visible: g_w=5, c_w=12 has 60 GHz shift but the shallowest dip (-10.8 dB), while g_w=3, c_w=28 has the deepest dip (-12.9 dB) but smallest shift (40 GHz).
- The ideal configuration would combine deep dip AND large shift - this is the multi-objective optimisation challenge that the ML model will address.

5. Key Findings and Interpretation

Parameter Sensitivity Ranking

Parameter	Dip Sensitivity (dB/um)	Freq Sensitivity (GHz/um)	Impact
dx (unit cell size)	0.117	11.7	Strongest effect
c_w (capacitor width)	0.040	8.8	Moderate effect
g_w (gap width)	Affects shift more than dip	-	Controls ON/OFF shift
w_au, w_au_top, w_graph, h_graph	Unknown - not yet varied	-	Need data

Key Conclusions from Phase 1

- **dx is the most influential parameter:** It has 3x the sensitivity of c_w on dip depth and also strongly controls resonance frequency. Larger dx values consistently give deeper dips.
- **g_w controls the ON/OFF shift:** g_w=5 gives 60 GHz shift while g_w=3 gives only 40 GHz. Wider gap allows graphene to have more influence on the resonance.
- **Current performance is far from target:** Best dip is -13.4 dB vs target of -20 dB (6.6 dB gap). This gap likely cannot be closed by varying only dx, g_w, and c_w - the unexplored parameters (w_au, w_au_top, w_graph, h_graph) must be investigated.
- **Multi-objective trade-off exists:** Deeper dips and larger frequency shifts partially conflict. The ML optimisation model will need to find the Pareto-optimal combinations.
- **Data is insufficient for ML:** Only 3 of 7 parameters have been varied, and only 13 simulations exist. At minimum 20-30 simulations with all parameters varied are needed to train a useful ML model.

6. Current Status - What Has Been Done So Far

- **Understanding the Problem:** Studied the COMSOL model, understood the 7 tuneable parameters, the S-parameter outputs, and the two optimisation objectives (deeper dip + larger frequency shift).
- **Data Pipeline Built:** Created data_loader.py that automatically parses COMSOL export files, extracts parameters from filenames, reads S-parameter curves, finds dips, and pairs ON/OFF states. This pipeline scales automatically as new data files are added.
- **Visualisation Tool Built:** Created visualize.py that generates 5 analysis plots examining parameter effects from different perspectives. All plots are saved as high-resolution PNGs.
- **Analysis Complete:** Created analyze.py that computes summary statistics, parameter sensitivity, identifies best configurations, flags missing data, and generates recommendations.
- **Initial Findings:** Identified dx as the most influential parameter, confirmed that g_w=5 gives better ON/OFF shift, and established that current data is insufficient for ML training.

7. Next Steps - What Will Be Done

Phase 2: ML Model Development

- Once 20-30 simulations are available, build Gaussian Process Regression model (ideal for small datasets with uncertainty quantification).
- Build Random Forest and XGBoost baseline models for comparison.
- Implement cross-validation to ensure models generalise well despite small dataset.
- Evaluate models using R-squared, RMSE, and MAE metrics.

Phase 3: Optimisation

- Implement Bayesian Optimisation to intelligently suggest promising parameter combinations.
- Build multi-objective optimisation (NSGA-II) to find Pareto-optimal trade-offs between dip depth and frequency shift.
- Generate ranked list of top 10 recommended parameter combinations.